UNITED INTERNATIONAL UNIVERSITY

Department of Computer Science and Engineering (CSE)

Course Syllabus

1	**Course Title**	Structured Programming Language Laboratory
2	**Course Code**	CSE 1112
3	**Trimester and Year**	Fall 2026
4	**Pre-requisites**	-
5	**Credit Hours**	1.0
6	**Section**	
7	**Class Hours**	
8	**Class Room**	
9	**Instructor's Name**	Ashraful Islam Paran
10	**Email**	paran@cse.uiu.ac.bd
11	**Office**	319(A)
12	**Counselling Hours**	Check Ucam
13	**Text Book**	1. Programming in ANSI C.(5th Edition or later) E. Balagurusamy 2. Teach Yourself C (3rd Edition or later) Herbert Schildt
14	**Reference**	1. Let Us C – Yaswant Kanetkar 2. Programming in C - Dennis Ritchie.
15	**Course Contents (approved by UGC)**	
16	**Course Outcomes (COs)**	
17	**Teaching Methods**	Lecture, Laboratory exercise and reports, Project Developments.

COs	Description
CO1	Describe the Structured Programming Language Features.
CO2	Design, compile and execute software programs to solve basic to moderate problems using a popular IDE.
CO3	Implementation of modular programs using different data structures like array, structures.
CO4	Work in a team and organize a project using abstraction through Structures.

18 CO with Assessment Methods

CO	Assessment Method	(%)
-	Attendance	5%
CO1, CO2, CO3	Coding Tests	30%
CO1, CO2, CO3	Home Assignment	15%
CO4	Viva	10%
CO1, CO2	Mid Exam	20%
CO1, CO2, CO3	Final Exam	20%

19 Mapping of Cos and Program outcomes

Cos	Program Outcomes(Pos)											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	C											
CO2		C										
CO3		C										
CO4			C						A	A		

20 Lab Outline

Class	Topics/Assignments	COs	Lab Outcomes/Activities
Lab1	Variable Declaration, Input-Output, Arithmetic expression evaluation, If-Else statement, Nested if-else, Switch-Case, Loops: for, while, do-while. (Recap of ICS Lab (CSE 1110))	CO1, CO2	Lecture, Coding Test, Assignment
Lab2	Advanced 1D loop problems, Usage of break and continue	CO1, CO3	Lecture, Coding Test, Assignment
Lab3	Advanced problems related nested loop (Pattern Printing etc.)	CO1, CO3	Lecture, Coding Test, Assignment
Lab4	Coding Test - 01		
Lab4	Declaration of 1-D array, Initialization of 1-D array, Simple programs related array	CO1, CO3	Lecture, Coding Test, Assignment
Lab5	Declaration of 2-D arrays, Initialization of 2-D arrays, Usage of 2-D arrays to solve different problems (i.e., matrices)	CO1, CO3	Lecture, Coding Test, Assignment
Lab6	Mid Exam		
Lab7	Emphasize the need for user-defined function, Definition of functions,	CO1, CO3	Lecture, Coding Test, Assignment

			Return values and their types, Arguments and their types, Function call, Function declaration, Function that returns multiple values, Nesting of functions, The scope of variables		
		Lab8	Declaring and initializing string variables, Reading string from terminals. Advanced problems related string.	CO1, CO3	Lecture, Coding Test, Assignment
		Lab9	Defining a structure, Declaring a structure, Accessing structure members, Structure initialization, Copying and comparing structure variables, Operations of individual members, Array of structure, Introduction to file handling	CO1, CO3, CO4	Lecture, Coding Test, Assignment
		Lab10	Coding Test - 02		
		Lab11	Pointers, Header files	CO1, CO3	Lecture, Coding Test, Assignment
		Lab12	Final Exam		

Appendix 1: Grading Policy

Letter Grade	Marks %	Grade Point	Letter Grade	Marks%	Grade Point
A (Plain)	90-100	4.00	C+ (Plus)	70-73	2.33
A- (Minus)	86-89	3.67	C (Plain)	66-69	2.00
B+ (Plus)	82-85	3.33	C- (Minus)	62-65	1.67
B (Plain)	78-81	3.00	D+ (Plus)	58-61	1.33
B- (Minus)	74-77	2.67	D (Plain)	55-57	1.00
			F (Fail)	<55	0.00

Appendix 2: Program outcomes

POs	Program Outcomes
PO1	An ability to apply knowledge of mathematics, science, and engineering
PO2	An ability to identify, formulate, and solve complex engineering problems
PO3	An ability to design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations
PO4	An ability to investigate complex problems using research-based knowledge and research methods design and conduct experiments, as well as to analyze and interpret data
PO5	An ability to use the techniques, skills, and modern engineering tools necessary for engineering practice
PO6	The broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context

PO7	Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex engineering problems in societal and environmental contexts
PO8	An understanding of professional and ethical responsibility
PO9	An ability function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings
PO10	An ability to communicate effectively
PO11	Project management and finance
PO12	A recognition of the need for, and an ability to engage in life-long learning